

Solubility and Colligative Properties

Molality, freezing and boiling points, osmotic pressure

Name _____ Class _____ Date _____

EQUATIONS AND DATA

$$m = \text{mol solute} \div \text{kg solvent} \cdot \Delta T_f = iK_f m \cdot \Delta T_b = iK_b m \cdot \Pi = iMRT$$

$$\text{Water: } K_f = 1.86 \text{ }^\circ\text{C kg/mol} \cdot K_b = 0.512 \text{ }^\circ\text{C kg/mol} \cdot R = 0.08206 \text{ L atm mol}^{-1} \text{ K}^{-1} \cdot T(\text{K}) = t(^{\circ}\text{C}) + 273.15 \cdot 1 \text{ atm} = 760 \text{ torr}$$

$$\text{Molar masses (g/mol): glucose } 180.16 \cdot \text{CaCl}_2 110.98 \cdot \text{NaCl } 58.44$$

Show units on every quantity in every step.

A. Understanding

1. Define (a) colligative property (b) molality (c) van 't Hoff factor. (d) Why does a solution boil at a higher temperature than the pure solvent?

2. Explain: (a) why sugar dissolves in water but not in hexane (b) why fish may struggle in warm water (c) why a fizzy drink bubbles when the cap is removed.

B. Freezing and boiling points

3. 18.0 g of glucose is dissolved in 100. g of water. Find (a) the molality (b) the freezing point (c) the boiling point.

4. Put these 0.10 mol/kg aqueous solutions in order of freezing point, highest first: CaCl_2 , glucose, NaCl. Explain.

5. 11.1 g of CaCl_2 is dissolved in 250. g of water. Find the freezing point ($i = 3$).

6. 3.00 g of a molecular compound dissolved in 50.0 g of water raises the boiling point by 0.256°C . Find its molar mass.

C. Osmotic pressure

7. 1.00 g of a protein is dissolved in water to make 100.0 mL of solution. Its osmotic pressure at 25.0 °C is 3.80 torr. Find the molar mass of the protein.

8. Blood has an osmotic pressure of about 7.70 atm at 37.0 °C. What concentration of NaCl ($i = 2$) would have the same osmotic pressure?

D. Challenge

9. What mass of NaCl must be dissolved in 1.00 kg of water to lower its freezing point to -10.0 °C ($i = 2$)? Suggest why calcium chloride is often used instead in very cold places.